

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		$\text{CH}_3-\text{C}(=\text{O})\text{OCH}_3 + \text{NaOH} \longrightarrow \text{CH}_3-\text{C}(=\text{O})\text{O}^-\text{Na}^+ + \text{CH}_3\text{OH}$		1		1		
		(ii)	I	88		1		1	1	
			II	$\text{H}-\text{C}(=\text{O})\text{OCH}_2\text{CH}_2\text{CH}_3 \quad \text{or} \quad \text{H}-\text{C}(=\text{O})\text{OCH}(\text{CH}_3)_2 \quad (1)$ ester must contain $\text{H}-\text{C}(=\text{O})$ group for it to reduce Tollens reagent (1)		1	1	2		
		(iii)	I	60		1		1	1	
			II	if relative molecular masses are different, the number of moles present in the chromatogram is not directly related to the mass of each component			1	1		
		(iv)		A has weaker intermolecular forces therefore (less energy is needed to overcome these forces and) it has a lower boiling temperature			1	1		
		(v)		2-methylbut-1-ene			1	1		
				Question 9 total	2	6	6	14	4	4